

Luigi Petrucco

AI Engineer & Neuroscientist

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🌐 vigji.github.io

🐙 [vigji](https://github.com/vigji)

From neural circuits to neural networks.

Experience

AI Engineering

- Sep 2025 – **Data Scientist & AI Engineer**, *Afliant S.r.L.*, Rovereto, Italy
present Developing and evaluating AI solutions for business clients; driving the company's adoption of AI-assisted software development.
- ML pipelines for data cleaning and forecasting model evaluation (Airflow / MLFlow).
 - Data platform for traffic data on Databricks: ingestion, cleaning, model training.
 - Synthetic multi-agent populations for marketing research; multi-agent flows for specification-driven development.
- Sep 2022 – **AI Engineering Consultant**, *ICOP S.p.A.*, Udine, Italy
present Embedded data and AI function within the technical organization of an Italian engineering and construction firm. *Concurrent role.*
- Built and evaluated MCP servers and RAG systems for an internal company chatbot.
 - Designed, evaluated and deployed multimodal LLM pipelines for delivery-note / invoice matching in the accounting process.
 - Python applications for data parsing and aggregation in financial consolidation.

AI Safety & Red-teaming

- 2026 **Red-teaming & adversarial robustness**, *Competitive leaderboards*
Independent work on jailbreaking, prompt injection, and the adversarial robustness of frontier LLMs and LLM agents.
- **Top 30** on the JudgmentDay red-teaming arena (AIM Intelligence).
 - **Top 50** in the Gray Swan indirect prompt-injection challenge.

Research

- Mar 2022 – **EMBO Postdoctoral Researcher**, *Istituto Italiano di Tecnologia*, Rovereto, Italy
Aug 2025 Biological bases of pursuit behavior.
- Deep learning pipelines for pose estimation from video; processing of large-scale electrophysiology datasets.
 - Hidden Markov models of animal behavior.
 - Supervision of MSc and PhD students (Neuroscience & Data Science).
- Sep 2017 – **Ph.D. Researcher**, *Max Planck Institute of Neurobiology / TUM*, München, Germany
Apr 2022 Ring attractor networks in the larval zebrafish brain. Identified a previously undescribed neural circuit acting as a heading compass. Thesis awarded **Best European PhD Thesis in Neuroscience 2022–2024** (FENS-Kavli).
- Time-series analysis on neural data: filtering, dimensionality reduction, clustering.
 - Processing of large-scale volumetric imaging data.

Open-source software

Sep 2020 – **Founder & core developer**, *Brainglobe*
 Apr 2022 Python ecosystem for computational neuroanatomy. **Received over \$400k in funding**; now maintained at the Sainsbury Wellcome Centre, London.
 ○ Core libraries: *brainglobe-atlasapi* (download, versioning and loading of volumes, meshes, ontologies); *brainglobe-space* (3D geometry for data transformation).
 ;Stytra — Python library for closed-loop virtual reality and image segmentation in animal experiments.

Teaching

Apr 2022 – **Lecturer, *Python for Neuroscience***, *Università di Trento*, Trento, Italy
 present Founded and teach a Python course for PhD students in Cognitive and Brain Sciences.

Education

2017–2022 **Ph.D. in Neuroscience**, *Ludwig-Maximilians-Universität (LMU) / Max Planck*, München, Germany
 FENS-Kavli prize for best European PhD thesis 2022–2024.
 2014–2016 **M.S. in Neuroscience**, *University of Pisa*, Pisa, Italy, 110/110 cum laude
 2011–2014 **B.S. in Biology**, *University of Pisa*, Pisa, Italy, 110/110 cum laude
 2011–2016 **Diploma**, *Scuola Normale Superiore*, Pisa, Italy
 Highly selective honors program; advanced courses in calculus, algebra, probability, and physics, taken in parallel with the BS/MS.

Skills

	Level	Skill	Years	Comment
Languages	■■■■■	Python	9	<i>Scientific stack, PyTorch, OpenCV, Qt</i>
	■■■□□	SQL / MongoDB	4	<i>PostgreSQL, MySQL</i>
	■■□□□	MATLAB	2	
ML / AI	■■■■■	PyTorch	4	<i>Training, distillation, mech-interp probes</i>
	■■■■■	LLMs & agents	2	<i>RAG, MCP servers, multimodal & agentic pipelines</i>
	■■■□□	Red-teaming	1	<i>Jailbreaks, prompt injection, robustness eval</i>
	■■■□□	HMMs	6	<i>Behavioral & neural time-series inference</i>
Data & MLOps	■■■□□	Airflow / MLFlow	1	<i>DAGs, experiment tracking, model registry</i>
	■■■□□	Databricks	1	<i>Ingestion, cleaning, model training</i>
	■■■□□	Docker / Compose	2	<i>Multi-container, self-hosted apps</i>
	■■■■■	GitHub Actions	9	<i>Testing, PyPI deploy, agentic CI flows</i>
Other	■■■■■	Adobe / Blender	8	<i>Graphic design enthusiast</i>

Awards & funding

- 2024 **Best European PhD Thesis in Neuroscience**, *FENS-Kavli Network*
Awarded for the 2022–2024 cycle.
- 2023 **EMBO Postdoctoral Fellowship**, *European Molecular Biology Organization*
Two-year postdoctoral funding (~€130k).
- 2022 **Talk invitation**, *CoSyNe 2022 (Computational and Systems Neuroscience)*
Abstract selected from the top 3%.
- 2011 **Scuola Normale Superiore Scholarship**
Full tuition and stipend.

Languages

Italian (native), English (proficient), German (beginner).

Selected writing

Personal blog at vigji.github.io — essays on AI-assisted software development, mechanistic interpretability, and complex systems.

Selected publications

Luigi Petrucco, Hagar Lavian, You Kure Wu, Fabian Svara, Vilim Štih, and Ruben Portugues. Neural dynamics and architecture of the heading direction circuit in zebrafish. *Nature neuroscience*, 26(5):765–773, 2023.

Vilim Štih, Luigi Petrucco, Andreas M Kist, and Ruben Portugues. Stytra: An open-source, integrated system for stimulation, tracking and closed-loop behavioral experiments. *PLoS computational biology*, 15(4):e1006699, 2019.

Federico Claudi, Luigi Petrucco, Adam Tyson, Tiago Branco, Troy Margrie, and Ruben Portugues. Brainglobe atlas api: a common interface for neuroanatomical atlases. *Journal of Open Source Software*, 5(54):2668–2668, 2020.

Federico Claudi, Adam L Tyson, Luigi Petrucco, Troy W Margrie, Ruben Portugues, and Tiago Branco. Visualizing anatomically registered data with brainrender. *Elife*, 10:e65751, 2021.